

ES200: ENVIRONMENTAL STUDIES - Science & Engineering**2024 Autumn Semester****Final Exam****(Max. Marks: 100, Time: 120 minutes)**

Note: There is no negative marking for multiple-choice questions.

Please fill in your response to each question in the matrix below against respective question numbers

ANSWER KEY

1) cX	5) d✓	9) a✓	13) b✓	17) d✓	21) b✓	25) a✓	29) d✓	33) c✓	37) a✓
2) cX	6) a✓	10) c✓	14) c✓	18) d✓	22) b✓	26) a✓	30) c✓	34) b✓	38) b✓
3) d✓	7) c✓	11) a✓	15) b✓	19) a✓	23) b✓	27) c✓	31) d✓	35) d✓	39) cX
4) a✓	8) c✓	12) d✓	16) c✓	20) b✓	24) a✓	28) b✓	32) a✓	36) c✓	40) b✓

- Emission factors are used to estimate the emissions from a source based on the level of activity as shown in the examples below. Select the correct answer:
 - Line sources in g/km
 - Coal power plants in mg/kW
 - ☒ Playground area dust sources in kg/(km².s)
 - All of the above
- If the moisture content on dry basis is 15%, what is the moisture content on wet basis?
 - 13.04%
 - 17.64%
 - ☒ 10%
 - 21%
- Organic pollution in a water body largely affects its
 - pH value
 - alkalinity
 - temperature
 - ☒ dissolved oxygen
- Which is the correct sequence of waste hierarchy (starting from most preferred one):

- a. Reduction, Recycle, Composting, Energy recovery, Landfilling
 - b. Reduction, Composting, Recycle, Energy recovery, Landfilling
 - c. Reduction, Recycle, Energy recovery, Composting, Landfilling
 - d. Reduction, Composting, Recycle, Landfilling, Energy recovery
5. Understanding the cause and effect in air pollution requires several steps. Indicate which of the following includes the key steps to provide a course of actions for the decision makers:
- a. Source apportionment, receptor modelling, health records, meteorology
 - b. Meteorology, road sweeping, monitoring, fogging at construction sites
 - c. Ban of garbage burning, ban of stubble burning, artificial rain, satellite data
 - d. Inventory, modelling, monitoring, source control
6. Air pollutants are classified as Primary and Secondary because:
- a. Primary are emitted from a source while secondary are formed in the atmosphere
 - b. Secondary are emitted from a source while primary are formed in the atmosphere
 - c. Neither primary nor secondary are emitted directly from a chimney
 - d. Primary are from tall chimneys and disperse easily while secondary are from short chimneys and therefore more local
7. Why is the PM_{2.5} to PM₁₀ ratio important?
- a. PM_{2.5} represents only the respirable fraction of PM₁₀
 - b. If it is greater than 1, it is harmful
 - c. It represents the anthropogenic fraction of PM
 - d. PM mass greater than 2.5 μm is toxic
8. Criteria pollutants are used in air quality to indicate:
- a. Sources of particulate pollutants
 - b. Sources of gaseous pollutants
 - c. Sources of pollutants associated with human activities
 - d. Sources of secondary sources
9. Global warming potential (GWP) for CO₂ is
- a. 1
 - b. 100
 - c. 0
 - d. Not defined
10. In India, the major use of groundwater resources is for
- a. Domestic activities
 - b. Industrial activities
 - c. Agricultural activities
 - d. Commercial activities

11. In reverse osmosis, the applied pressure in the process is
- ☒ a. $>$ osmotic pressure
 - b. $<$ osmotic pressure
 - c. Equal to osmotic pressure
 - d. Osmotic pressure has no bearing on the applied pressure
12. Which of the following is not a step during a conventional activated sludge process in secondary treatment for wastewater treatment plant operation?
- ☒ a. Aeration
 - b. Recycle of sludge from secondary clarifier
 - c. Settling of sludge
 - d. Sludge digestion
13. Face masks for personal protection use filters. The three key mechanisms of removing particles in a filter are:
- a. interception, thermophoresis and diffusiophoresis
 - ☒ b. Diffusion, interception and impaction
 - c. Sieving, inertia and electrostatic precipitation
 - d. Absorption, adsorption and deliquescence
14. Which of the following screens is also called 'Trommel' screen?
- a. Disc
 - b. Vibrating
 - ☒ c. Rotating
 - d. Reciprocating
15. Gasification of solid waste is performed with.....
- ☒ a. No air
 - b. starved air
 - c. with stoichiometric air
 - d. with excess air
16. Which of the following is not true?
- a. Typically, rain is diluted version of seawater
 - b. The key driving force for the hydrological cycle on earth is sun
 - ☒ c. Typically, rainwater has higher alkalinity than river water.
 - d. Typically, rainwater is acidic ($\text{pH} < 7$).
17. If the CO_3^{2-} ions concentration in a water sample is 90 mg/L and pH of water sample is 10.3. Find the alkalinity of the sample. Take acid dissociation constants $K_{a,1}$ and $K_{a,2}$ for H_2CO_3 as $10^{-6.3}$ and $10^{-10.3}$ respectively.
- a. 225 mg/L as CaCO_3
 - b. 250.5 mg/L as CaCO_3
 - c. 275.5 mg/L as CaCO_3
 - ☒ d. 234.5 mg/L as CaCO_3

18. Arrange various waste constituents in decreasing order of biodegradability (starting with material having highest biodegradability):

- a. Cardboard, Newspaper, Food waste, Plastics,
- b. Food waste, Newspaper, Plastics, Cardboard
- c. Food waste, Newspaper, Cardboard, Plastics
- d. Food waste, Cardboard, Newspaper, Plastics

0.82 0.47 0.35

19. High purity oxygenation in an aeration tank during conventional sludge process will

- a. reduce the residence time of aeration
- b. reduce the required volume of the aeration tank
- c. Both a and b
- d. Neither a nor b

20. The chemical formula of a municipal solid waste sample is $C_{90}H_{220}O_{100}N$. The energy content of waste material will beMJ/kg.

Use the following if needed:

Heating value (kJ/kg) = $337C + 1428(H - 0.125O) + 93S + 23N$

Atomic mass: C = 12, H = 1, O = 16, N = 14 and S = 32

- a. 11
- b. 13.6
- c. 15.2
- d. None of these

21. Schmutzdecke layer is formed in

- a. rapid sand filtration
- b. slow sand filtration
- c. activated carbon packed bed
- d. All of the above

22. Comparison of actual (environmental) lapse rates and adiabatic lapse rate is used to determine conditions of stability in the atmosphere for the tendency for vertical mixing. What is the benefit of vertical mixing?

- a. Prevents horizontal transport to other regions
- b. Spreads the pollutants into a larger volume
- c. Keeps the pollution away from breathing level
- d. Makes it possible for moisture to absorb particulate matter

23. The pollutant most widely used to study changes in the atmosphere as an indicator of climate change:

- a. Water
- b. CO_2
- c. SO_2
- d. $PM_{2.5}$

24. In air pollution dispersion modelling, air pollution sources are classified into three broad groups as:
- ☒ a. Area, Line and Point sources
 - b. Industrial, Domestic and Forest fires
 - c. Two wheelers, Cars and Trucks
 - d. Organic emissions, Inorganic emissions and Particulate emissions
25. Biogas is a mixture of.....
- ☒ a. $\text{CH}_4 + \text{CO}_2$
 - b. $\text{CO} + \text{H}_2$
 - c. $\text{CH}_4 + \text{H}_2$
 - d. $\text{H}_2 + \text{CO}_2$
26. During vermi-composting, 1 kg of worms can consume.....kg of waste:
- ☒ a. 1
 - b. 1.5
 - c. 2
 - d. 2.5
27. Which of the following is NOT a source of grey water from homes?
- a. Laundry
 - b. Kitchen tap
 - ☒ c. Toilet
 - d. both (b) & (c)
28. Stockholm Convention deals with controlling which of the following contaminants/class of contaminants?
- a. Mercury
 - ☒ b. Persistent organic pollutants
 - c. Microplastics
 - d. Pathogens
29. Meteorologists use a graphical representation called Wind-Rose. It indicates:
- a. Height of the smoke plume, Mean Sea Level, and the Latitude
 - b. Rising of the wind, Temperature and Pressure gradient
 - c. Wind Velocity, Humidity, and Temperature
 - ☒ d. Speed of the wind, direction, and % of time
30. As per SWM rules (2016), at least.....years post-closure care of landfills is needed:
- a. 5
 - b. 10
 - ☒ c. 15
 - d. None of these

31. With respect to their increasing pore sizes, the membranes can be arranged as (low pore size to high pore size)
- a. NF, UF, MF, RO
 - b. UF, MF, NF, RO
 - c. RO, NF, MF, UF
 - d. RO, NF, UF, MF
32. Which of the following is not true for ISO 14001 Standard?
- a. It is related to specific environmental performance criteria
 - b. It can be used in whole or in part to improve environmental management
 - c. It is applicable to any organization regardless of their size, type and nature.
 - d. It specifies the requirements for an environmental management system
33. How would one know for sure if the air in your city is polluted?
- a. Black dirt on your shirt collars
 - b. Visibility in winter in a hill station like Darjeeling
 - c. By comparing with NAAQS
 - d. Organic smell at some times during a day
34. Gaseous emissions are controlled in the industry using:
- a. Cyclones, adsorption, incineration
 - b. Adsorption, incineration, absorption
 - c. Fabric filters, incineration, catalytic conversion
 - d. Impactors, electrostatic precipitators, incineration
35. Particulate emissions are controlled in the industry using:
- a. Cyclones, adsorption, incineration
 - b. Adsorption, incineration, absorption
 - c. Fabric filters, incineration, catalytic conversion
 - d. Cyclones, electrostatic precipitators, fabric filters
36. A perc test is conducted in a soil to determine feasibility of septic tank. During the test, it was observed that the water level was dropped by 150 mm in 30 min duration. The 'Perc rate' will be.....
- a. 5 mm/min
 - b. 0.2 mm/min
 - c. 0.2 min/mm
 - d. 5 min/mm
37. Sewage is to be aerated in an activated sludge tank (length = 30 m, width = 10 m and water depth = 4 m). The sewage flow rate is 4 Million liters per day. The effluent from primary clarifier has a 5-day BOD of 130 mg/l. The MLSS is to be kept at 2000 mg/l. The food to microorganism ratio will be
- a. 0.22 d^{-1}

- b. 0.87 d^{-1}
- c. 51 d^{-1}
- d. None of these

38. Which of the following is considered 'preliminary treatment' in a sewage treatment plant?

- a. Primary sedimentation tank
- ☒ b. Grit chamber
- c. Chemical precipitation
- d. both a & b

39. Ozone is :

- a. A primary pollutant
- b. A secondary pollutant
- ☒ c. Not a criteria pollutant
- d. Harmless in the troposphere

40. Match the following

Water Quality Parameter	Measurement Approach
(A) Chemical oxygen demand	1.) Scattering of light in a nephelometer
(B) Biochemical oxygen demand	2.) Potassium dichromate titration
(C) Colour	3.) Spectrophotometric technique at 455 nm wavelength
(D) Turbidity	4.) Incubation with bacterial seed at 20°C for 5 days

- a. A-3, B-1, C-4, D-2
- ☒ b. A-2, B-4, C-3, D-1
- c. A-4, B-1, C-2, D-1
- d. A-1, B-2, C-3, D-4

Final Exam

(Max. Marks: 100, Time: 120 minutes)

Note: There is no negative marking for multiple-choice questions.

Please fill in your response to each question in the matrix below against respective question numbers

A

ANSWER KEY Section A

1) (d)	5) (d)	9) (a)	13) (b)	17) (d)	21) (b)	25) (a)	29) (d)	33) (c)	37) (a)
2) (a)	6) (a)	10) (c)	14) (c)	18) (d)	22) (b)	26) (a)	30) (c)	34) (b)	38) (b)
3) (d)	7) (c)	11) (a)	15) (b)	19) (c)	23) (b)	27) (c)	31) (d)	35) (d)	39) (b)
4) (a)	8) (c)	12) (d)	16) (c)	20) (b)	24) (a)	28) (b)	32) (a)	36) (c)	40) (b)

[Handwritten signature]